



GE Medical Systems

Technical Publications

Direction 2252520

Revision 3

Signa® Gradient Isolation Kit Installation

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Service Documentation

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation "**damage in shipment**" written on **all** copies of the freight or express bill **before** delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

Immediately complete a "Damage Loss Claim Form", available via MS Exchange Mail, after the damage is found.

MS Exchange Path:

Outlook/Public Folder/All Public Folders/Medical Systems/!Global Initiatives/Information Management/Forms/Common Forms/DAMAGE LOSS CLAIM FORM.

Send the completed form to the email address listed in the form.

For more information about the Transportation Claim Procedure, access the GE Medical Systems Intranet and enter the following URL address (case sensitive):

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Rev. 4/2/2007

Direction 2128126 - Language Policy For Service Documentation, Rev 0

WARNING

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.

AVERTISSEMENT

- CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
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WARNUNG

- DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH NICHT ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENTE REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTE AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

警告

- ・このサービスマニュアルは英語版しかありません。
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注意：

- 本维修手册仅存有英文本。
- 非 GEMS 公司的维修员要求非英文本的维修手册时，客户需自行负责翻译。
- 未详细阅读和完全了解本手册之前，不得进行维修。
- 忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

REVISION HISTORY

REV	DATE	PRIMARY REASON FOR CHANGE
0...	December 20, 1999	Initial Release.
1.....	April 13, 2001	Added steps to install radial supports in sections 3 & 4.
2.....	March 30 ,2007	Update for revised lead board Kits
3.....	June 12, 2009	Updated Gray Foam part number to 2379255-2 in Section 5, step 15. Added location of foam installation in illustration.

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1. PERSONNEL REQUIREMENTS

Required Persons	Preliminary Reqs	Procedure	Finalization
1	Timing Not Applicable	2 hours	Timing Not Applicable

2. TOOLS, TEST EQUIPMENT, KIT CONTENTS

2.1. Tools and Test Equipment

Coil Lift Tool Kit for Gradient Isolation Kit Part # 2261179 (Available from Tools Depot)

Description	Quantity	Part Number
Magnet Interface Bracket, Right	1	2256832
Magnet Interface Bracket, Left	1	2256833
Gradient Coil Interface Bracket	2	2256834
Bolt	2	2256835
Hex Head Cap Screw 35 mm M10x1.5 A2-70	4	2109866-22
Hex Head Cap Screw 100 mm M10x1.5 A2-70	8	2109866-27
Metric Standard Flat Washers M10 A2	12	2109878-4

2.2. List of Equipment & Parts

1. CRM Isolation Upgrade Install Kit Part # 2247493

Kit Contains:

Description	Quantity	Part Number
Support Mounting (Bracket)	1	2240708
Spacer Shim 1.5 mm THK	2	2238908
Spacer Shim 2.4 mm THK	6	2238908-2
Spacer Shim 16 mm THK	1	2238908-3
Rubber Isolation Pad	2	2241176
Set Screw 10 mm 20 mm Grub	2	2251294
BRM Isolation Upgrade Install Kit	1	2247493IDW
Spacer Shim 10 mm THK	1	2238908-4
CRM Rear Support Mounting	1	2246675
Endbell Plate Mounting Spacer	1	2235604
Bostik "Never-Seeze" Anti-seize compound	1	46-294151P8
Screw set screw 10 mm 25 mm Grub	1	2251295
Upper Foam Spacer	1	2274743
Lower Foam Spacer	1	2274744
Gradient Cable Bracket	1	2274745
Cap Screw Hexagon Head, 10 mm 1.5 x 40 mm	1	2109866-23
Foam Gray Textured surface, 116" LG	1	2379255
Filler Pad Conquest --Diecut Inserts	4	2181231-2
Set Screw M16	4	2293535
Mtg Pad, 1/4" Thick Neoprene	4	2131953-4
Loctite 271, 10cc Bottle	1	46-170684P2
Loctite 242, 10cc Bottle	1	46-294151P2
Metric Hex head, nylon nut M16	4	2325862
Open Foam Caulking Strip ~ 0.500 THK	2	2179397-3
Gradient Isolation Kit Instructions	1	2252520

2. BRM Isolation Upgrade Install Kit Part # 2242380

Kit Contains:

Description	Quantity	Part Number
Support Mounting	2	2240708
Spacer 16 mm	1	2238907
Spacer Shim 1.5mm Thk	2	2238908
Spacer Shim 2.4 mm Thk	6	2238908-2
Spacer Shim 16 mm Thk	1	2238908-3
Rubber Isolation Pad	2	2241176
Spacer 28 mm	1	2238907-2
Screw Cheese Head 10 mm 20 mm Nylon	1	2209823-3
Screw Cheese Head 10 mm 25 mm Nylon	2	2209823-4
Screw Cheese Head 10 mm 35 mm Nylon	1	2209823-5
LCC/CX Mag Install parts List	1	2242380IDW
Bostik "Never-Seeze" Anti-seeze compound	1	46-294151P8
Foam Gray Textured surface, 116" LG	1	2379255-2
Filler Pad Conquest --Diecut Inserts	4	2181231-2
Mtg Pad, 1/4" Thick Neoprene	4	2131953-5
Loctite 271, 10cc Bottle	1	46-294151P11
Loctite 242, 10cc Bottle	1	46-294151P2
G10 Fiberglass Flat washer	8	2370892
Screw Hex socket cap m10x30 A2-70	8	2109867-24
Screw Hex socket cap m10x40 A2-70	2	2109867-26
Gradient Isolation Kit Instructions	1	2252520
Open Foam Caulking Strip ~ 0.500 THK	2	2179397-3
Metric Hex head, nylon nut M16	4	2325862
Set Screw M16	4	2293535
LCC BRM Mounting Pad	4	2212715-3

3. Preliminary Requirements

3.1. Prerequisite Procedures

1. "As installed" performance data of system should be available prior to any hardware changes.
 - Full SPT
 - White Pixel/Spike Noise

If this information is not available perform the appropriate test / procedure before continuing with this modification.

NOTE: Before shutting down the power, read through the entire procedure. Make sure the carriage is in the proper position for the bridge removal procedure.

NOTE: The following lock out tag out procedures should apply in most cases. However, if you have a system type not covered in this section, proceed to the safety section of the appropriate service documentation and follow the Lock Out Tag Out procedures.

3.2. Power Down & Lock Out Tag Out Procedures



POSSIBLE PERSONAL INJURY! AVOID SERIOUS INJURY OR DEATH BY ELECTROCUTION. REMOVE POWER FROM THE ACGD HARDWARE BEFORE WORKING ON THE MODULE. LOCK AND TAG OUT POWER TO THE ACGD HARDWARE SYSTEM WALL BOX DISCONNECT BEFORE STARTING WORK.

1. Bring the PC software down.

LX (SGI based) system software shutdown.

- a. Toggle mouse control to the PC monitor using the hard key on the top, left side of the keyboard housing.
- b. From the PC monitor use the mouse to select Start → Shutdown → Shut down.

Bring the Signa software down.

- c. Toggle mouse control back to the Signa monitor using the hardkey on the top, left side of the keyboard.
- d. Single-click on the Toolbelt icon.
- e. Single-click on the System Shutdown softkey on the Service Desktop Manager.
- f. Select OK to confirm the shutdown.

2. Wait for the system to indicate on the monitor that it is safe to power off the computer before proceeding. This usually takes about 90 seconds before this message is seen.

Horizon (Genesis console) system software shutdown.

Bring the Signa software down.

- a. Touch Utilities.
- b. Touch System Shutdown.
- c. Select OK to confirm the shutdown.
- d. Wait for Boot Prompt (this indicates it is safe to proceed).

3. Remove the cover from the front of the ACGD cabinet, See Illustration 2-3-1.



BREAKER LOCATION

ILLUSTRATION 2-3-1

4. Rotate the main breaker on the ACGD/PDU labeled INPUT counter-clockwise to the green, O (OFF) position. See Illustration 2-3-2.



MAIN INPUT BREAKER IN THE OFF POSITION

ILLUSTRATION 2-3-2

5. Verify that the 3 phase indicator lamps on the front of the power distribution unit (Illustration 2-3-5) are not illuminated

6. Lock Out / Tag Out the PDU "INPUT" circuit breaker.
 - a. Press on the left-pointing arrow on the end of the handle to expose the location where the lock and tag can be placed.
 - b. Lock and tag out the PDU "INPUT" circuit breaker as shown in Illustration 2-3-3



PDU MAIN INPUT CIRCUIT BREAKER LOCKED AND TAGGED OUT
ILLUSTRATION 2-3-3

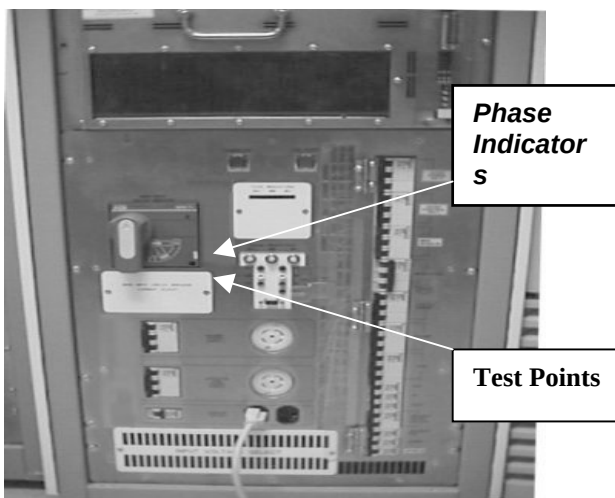


ILLUSTRATION 2-3-4

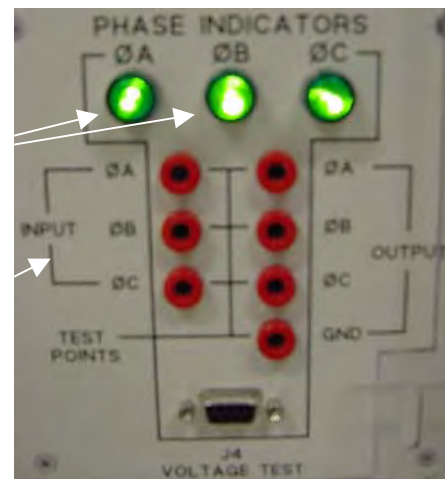


ILLUSTRATION 2-3-5

ACGD Cabinet –PDU (Front Cover Removed)

7. Verify that all energy has been dissipated. Measure the power at the front panel 420 and 208 test points of the PDUs. See Illustrations 2-3-4 & 2-3-5 voltages should now be at 0vac

Note: The test points are 100 :1 ratio (208 Vac = 2.08 Vac 420Vac = 4.20Vac)

3.3. Preliminary Disassembly Procedures

Locate the following procedures on the service document CD for the product. Perform the procedures in the order that they appear.

Signa Horizon Enclosure

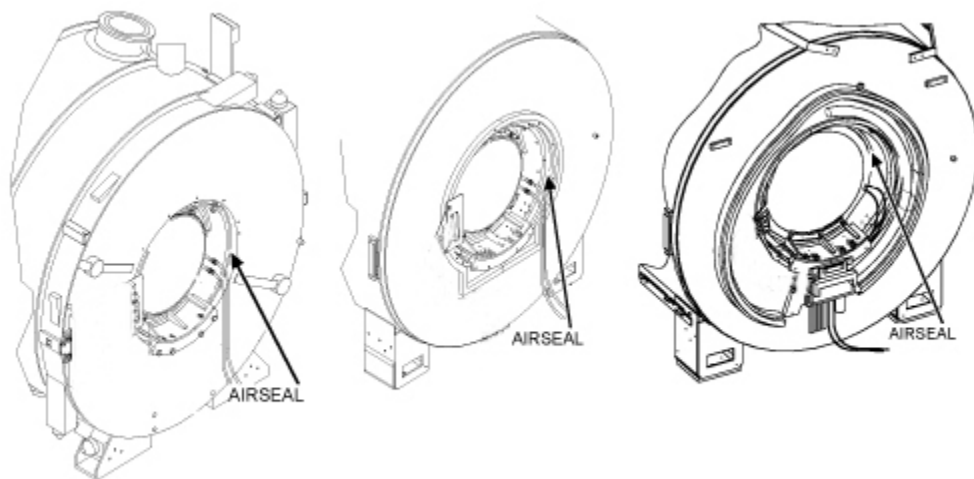
1. Rear Enclosure Cover Removal
2. Front Enclosure Cover Removal
3. Patient Comfort Module Removal
4. Bridge and Cradle Removal
5. Rear Pedestal Removal

Signa MR/i & CV/i WideOpen Enclosure

1. Top, Bottom Trim , & Side Arc Cover Removal
2. Rear Pedestal Cover Removal
3. Bridge and Cradle Removal
4. Rear Pedestal Removal
5. Rear Endbell Removal
6. Front Endbell Removal

4. Installation of (Front) Patient End Support Bracket

1. Remove air seal from the BRM/BRM-D/CRM. See Illustration 1.



CX MAGNET

LCC MAGNET

WIDEOPEN ENCLOSURE

REMOVE AIRSEAL ILLUSTRATION 1

2. Remove the Terminal Block or cable bracket located on the rear flange of the magnet.
See Illustration 2. (The Terminal Block is not used with the BRM-D and MR/i & CV/i WideOpen systems)



Note : If you have the terminal block on the left of this Illustration It must be removed. LCC Magnets with serial R001-R0200 and all CX magnets

**REMOVE TERMINAL BLOCK
ILLUSTRATION 2**

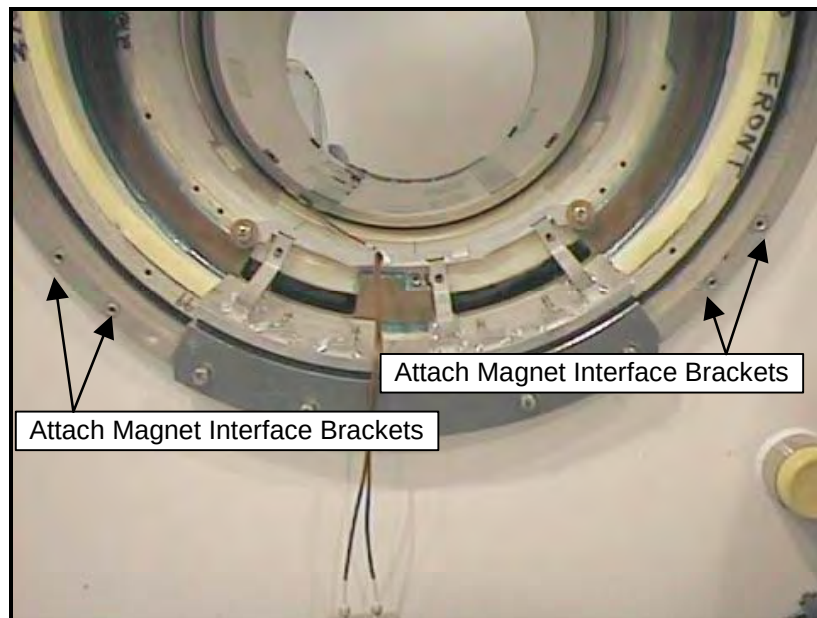
3. Loosen and remove the BRM/BRM-D/CRM M16 Hex Head Cap screws and radial support blocks (Mounting Pads) from both ends (2 per end) of the BRM/BRM-D/CRM Assembly. (See Illustration 8) Tool to use: 24 mm Non-Magnetic socket and ratchet. (Note: Do Not reuse M16 Stainless steel hex head cap screws: Replace with 2293535 set screw M16)
4. Add alignment marks to show the installed vertical and lateral alignment of each end of the gradient with respect to the cryostat. Alignment marks will be needed later during reassembly to Center the Coil and achieve equal gap between the magnet warm bore and the outer diameter of the gradient Coil (Left, Right and Top).
5. Remove 4 (four) M10 hex head bolts securing aluminum support brackets to magnet on the patient end and 4 (four) M10 bolts on the service end of the magnet (8 bolts total). Do not lift gradient coil with any of the 8 bolts in place.

6. Remove lower left and right threaded studs to allow for proper fit of the magnet interface brackets (See Illustration 3 and 4).

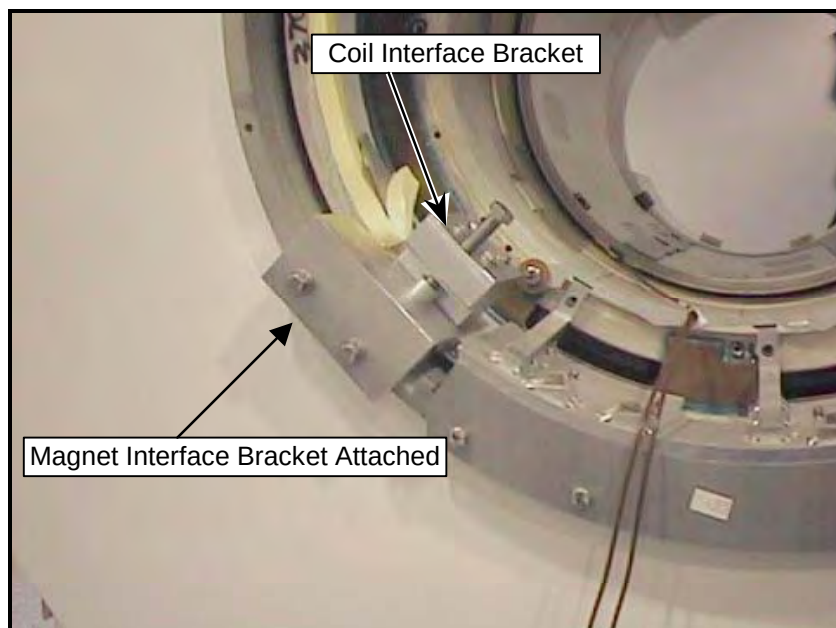
NOTE Property Damage! To avoid damage to the Magnet USE ANTI-SEEZE on the screws attaching the magnet interface brackets. The screws are made of Stainless Steel, which may cause the threads to gall and may result in cutting the magnet interface brackets off of the end flange.

NOTE Make sure the correct orientation is used when attaching the magnet interface brackets. For the CRM the side with the short ledge is attached on the gradient coil as marked on the bracket.

7. Install the left and right magnet interface brackets on the end flange using the 4 - 35 mm screws. See Illustration 3 and Illustration 4.



PLACEMENT OF MAGNET INTERFACE BRACKETS
ILLUSTRATION 3



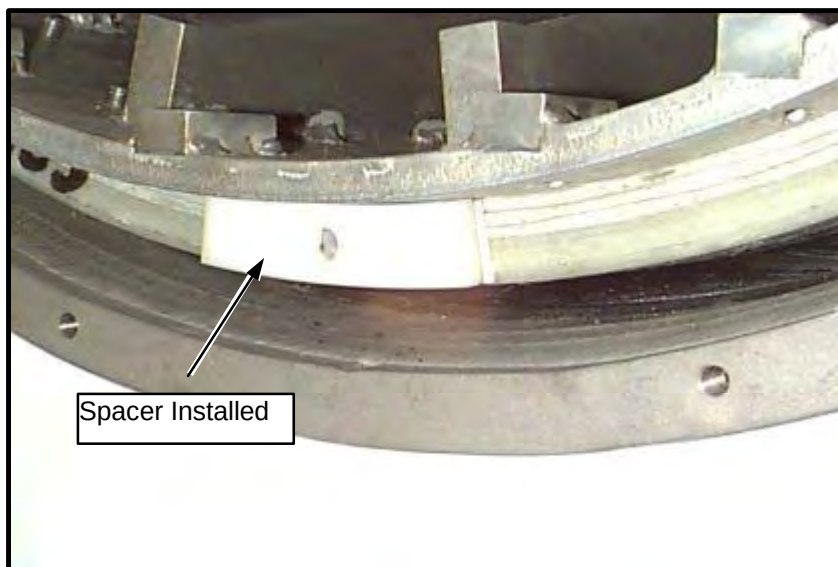
**MAGNET INTERFACE BRACKETS ATTACHED ON BRM
ILLUSTRATION 4**

8. Insert one bolt into each of the two coil interface brackets. See Illustration 5



**BOLT PLACED IN COIL INTERFACE BRACKET
ILLUSTRATION 5**

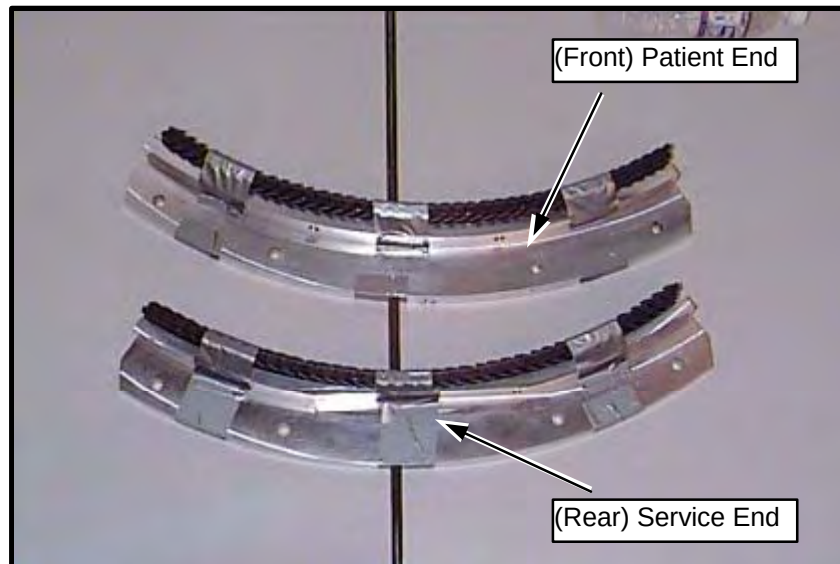
9. Remove poron seal gray pads from coil to expose the 4 bolt holes (8 holes total).
10. Attach both of the Coil Interface Brackets on the gradient coil using the 100 mm screws.
See Illustration 4.
11. Lift patient end of the gradient by tightening the bolt in the Coil Interface Bracket until the aluminum support brackets separate by at least 4 mm (1/8 inch).
12. Remove gradient coil support bracket from the patient end of the cryostat.
13. Install the fixed spacer (P/N 2238908-4 10 mm for CRM, P/N 2238907-2 28 mm for BRM) as shown in Illustration 6. Use the M10x25 nylon screws for BRM and CRM. The M10x25 nylon screw must be either level (flush) or inside the spacer hole. See Illustration 5.



**GRADIENT COIL LIFTED
ILLUSTRATION 5**

14. Refer to Gradient Placement Illustration 14 for proper configuration.
15. Install rubber pads (2241176) on top of aluminum bracket.

16. Install (front) patient end gradient coil support bracket. Use #242 blue loctite on the quantity 4 (2109867-24) hex screws. Make sure a G10 flat washers (2370892) are used on each of the hex screws (2109867-24) and secure to the end flange. See Illustration 6



**SUPPORT BRACKETS
ILLUSTRATION 6**

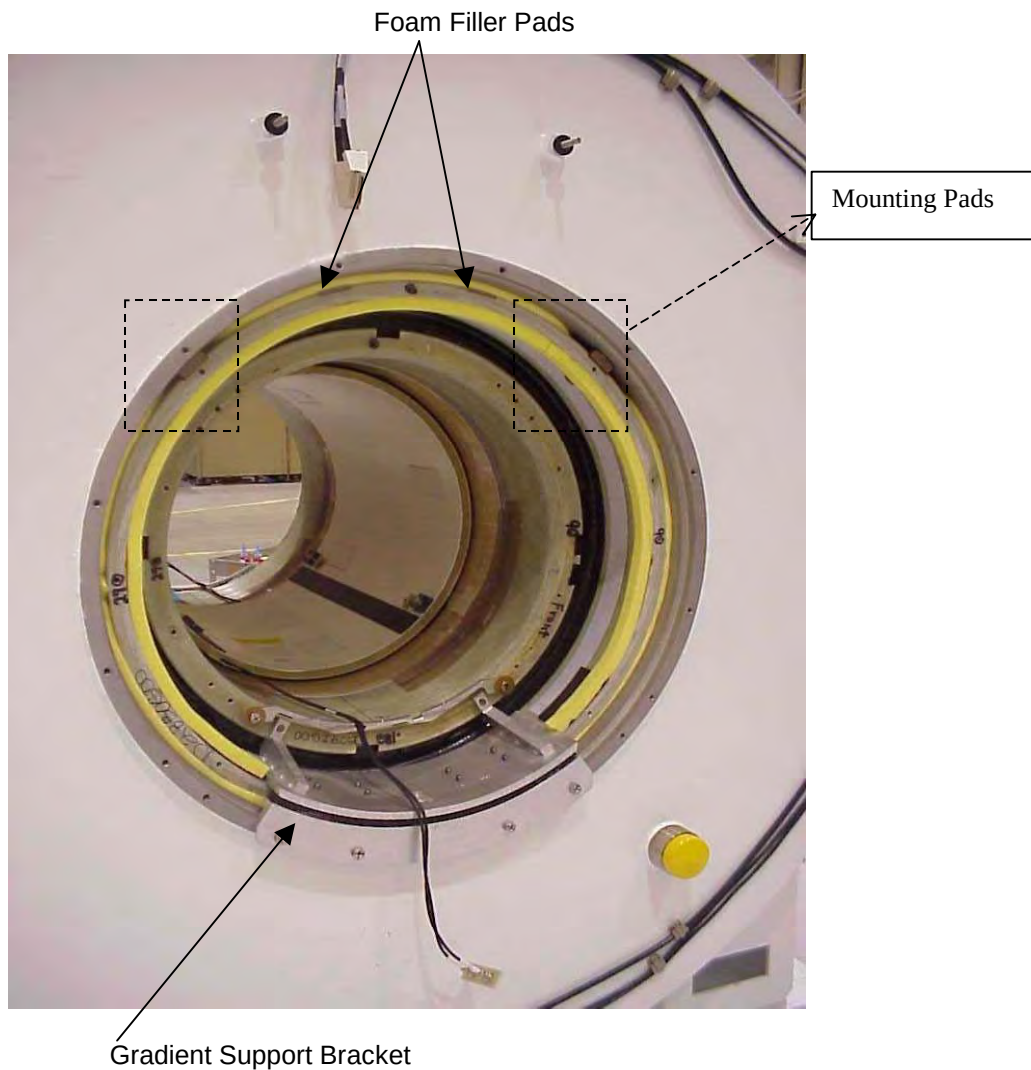
NOTE Do not use duct tape to install rubber pads. Duct tape was only used in picture to show rubber pads on brackets.

- o For CRM: bracket is installed with lip facing out. See Illustration 7.
- o For BRM: bracket is installed with lip under the existing bracket.



**SUPPORT BRACKET INSTALLED
ILLUSTRATION 7**

17. Lower patient end of gradient until weight is totally supported by new brackets.
18. Check alignment marks to verify that the gradient alignment was not altered, if altered, lift and adjust vertically to restore to original alignment
19. Remove Coil Interface Brackets and Right/Left Magnet Interface Brackets.
20. Install foam filler pads, part number 2181231-2 in top 2 pockets on gradient OD (outer diameter), see Illustration 8.



- **LOCATION OF FOAM FILLER PADS & UPPER RADIAL SUPPORTS**
ILLUSTRATION 8

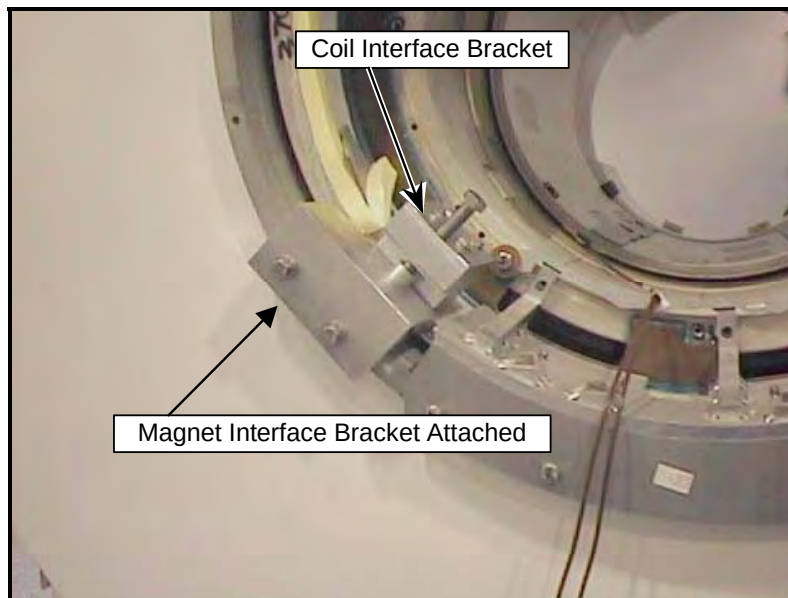
21. Pull out ½" black rubber cord from between gradient and magnet bore down to the gradient support bracket and discard.

5. Installation Of (Rear) Service End Support Bracket

1. Install the left and right magnet interface brackets on the end flange using the 4 - 35 mm screws. Illustration 9.

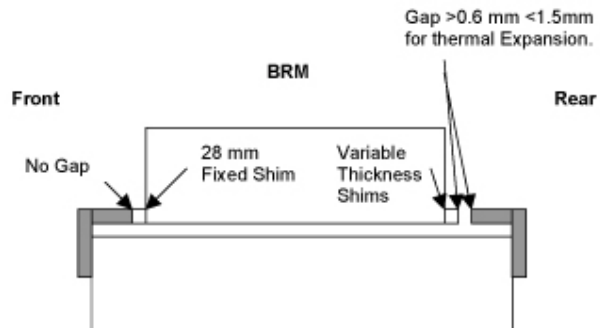
NOTE : Property Damage! To avoid damage to the Magnet USE ANTI-SEEZE on the screws attaching the magnet interface brackets. The screws are made of Stainless Steel, which may cause the threads to gall and may result in cutting the magnet interface brackets off of the end flange.

Illustration 9: Magnet Interface Brackets Attached On BRM



3. Remove the rear bottom RF Coil adjusting blocks to insure proper fit of the Coil Interface Bracket.
4. Attach both of the Coil Interface Brackets on the gradient coil using the 100 mm screws. See Illustration 9.
5. Lift service end of the gradient by tightening the bolt in each coil Interface bracket until the aluminum support brackets separate by at least 4 mm (1/8 inch).
6. Remove lower gradient support bracket from the service end of the cryostat.
7. Select a shim combination so the gap between the outermost shim and the support bracket is not more than 1.5mm and not less than 0.6mm. See Illustration 10

Illustration 10: Gradient Placement



8. Install the fixed spacer (the 16 mm [P/N 2238907] spacer and shims are required for the BRM) as shown in Illustration 11. Use the M10x25 nylon screws for BRM and CRM. Screw head must be either level (flush) or inside the spacer hole.

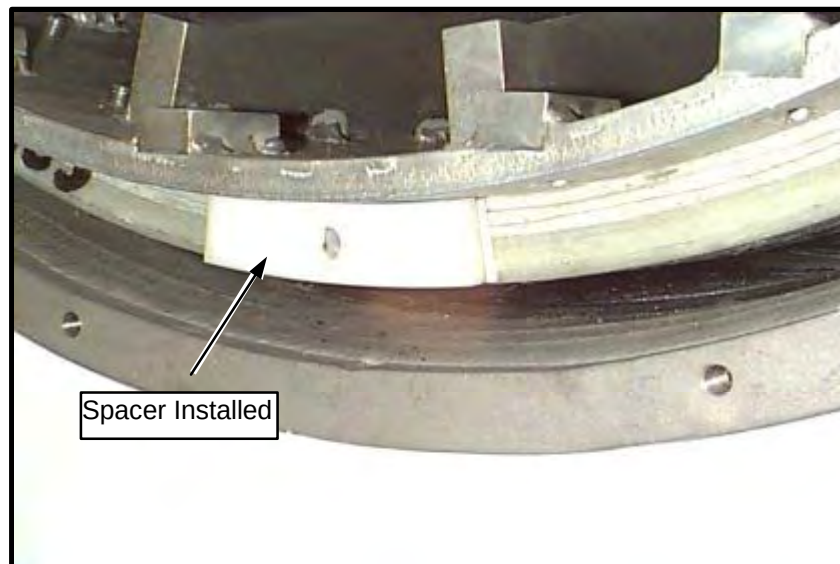


Illustration 11: Gradient Coil Lifted

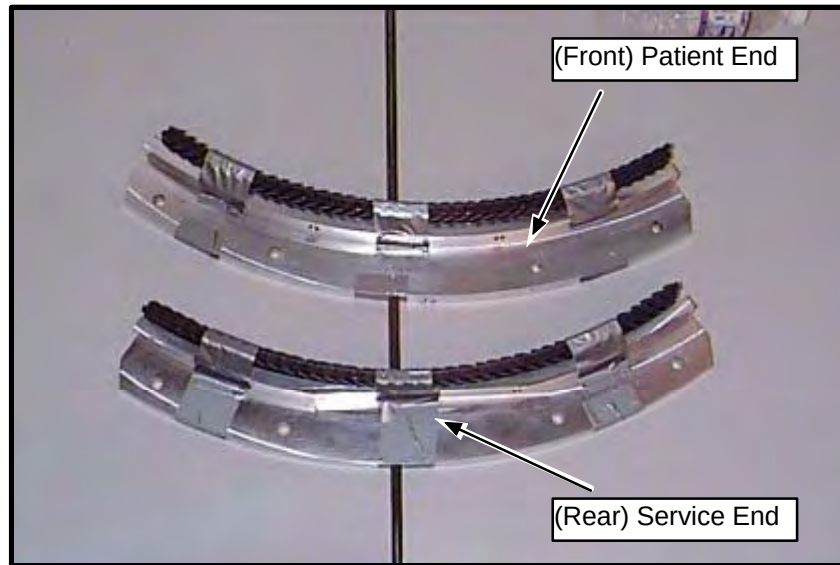
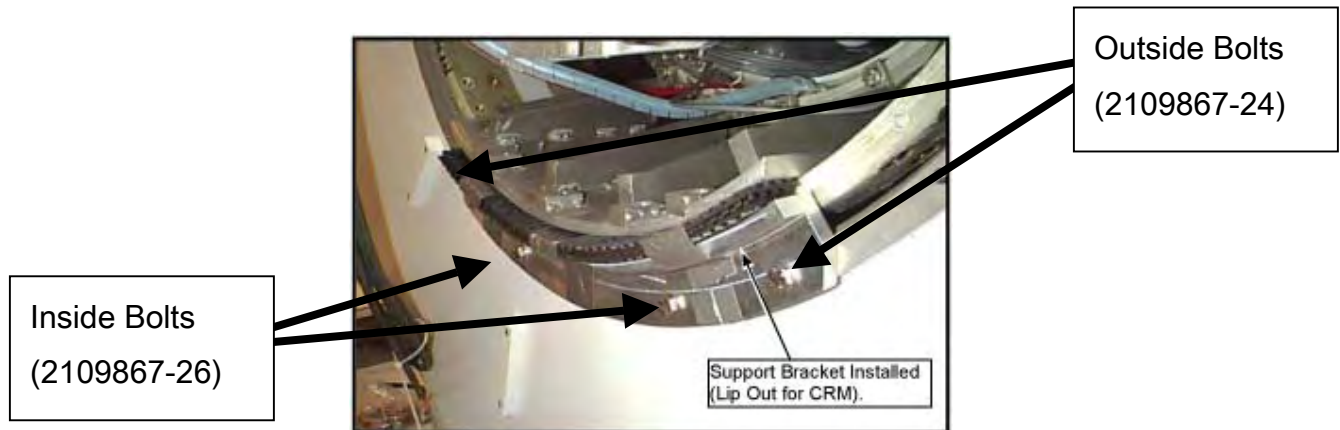


ILLUSTRATION 12: **SUPPORT BRACKETS SUPPORT BRACKETS**

9. Install (rear) service end bracket. Use #242 blue loctite on the screws between aluminum support brackets on the service end with quantity 2 (2109867-24) on the outside of the bracket and quantity 2 (2109867-26) on the inside of the bracket. Make sure a G10 flat washers (2370892) is used on each of the hex screws (2109867-24 & 2109867-26). See Illustration 12.
 - o For CRM: bracket is installed with lip facing out. See Illustration 13.
 - o For BRM: bracket is installed with lip under the existing bracket

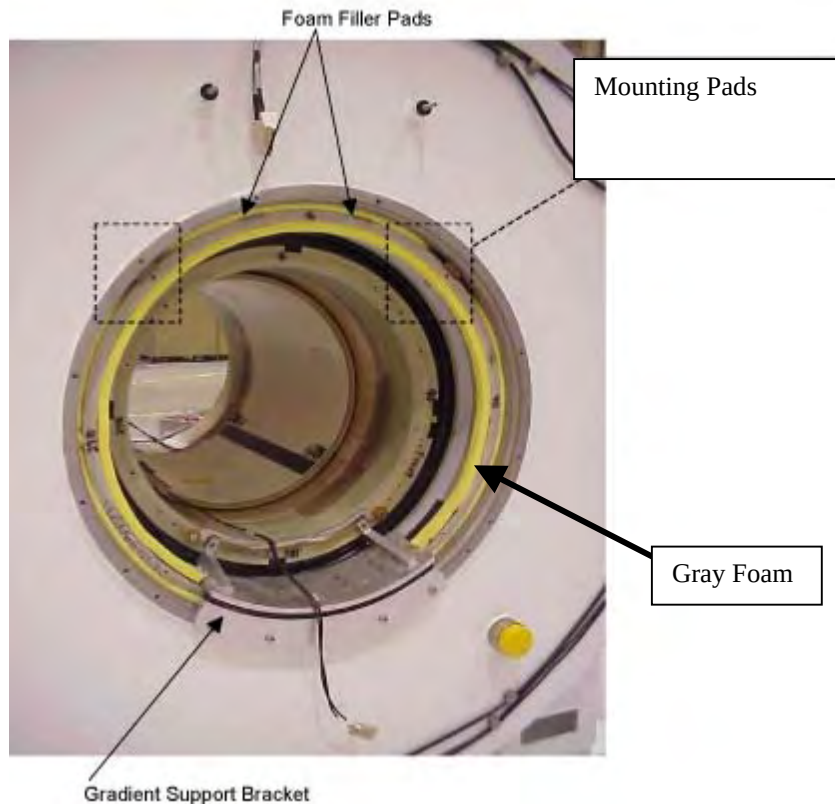
Illustration 13: Support Bracket Installed



10. Lower service end of gradient until weight is totally supported by new brackets. See Illustration 10.
11. Check alignment marks to verify that the gradient alignment was not altered, if altered, lift and adjust vertically to restore to original alignment.
12. Remove Coil Interface Brackets and Right/Left Magnet interface brackets.
13. Install foam filler pads, part number 2181231-2 in top 2 pockets on gradient OD (outer diameter), see Illustration 14.

NOTE The air seal requires a uniform gap between the coil outer diameter and inner diameter of the magnet warm : bore.

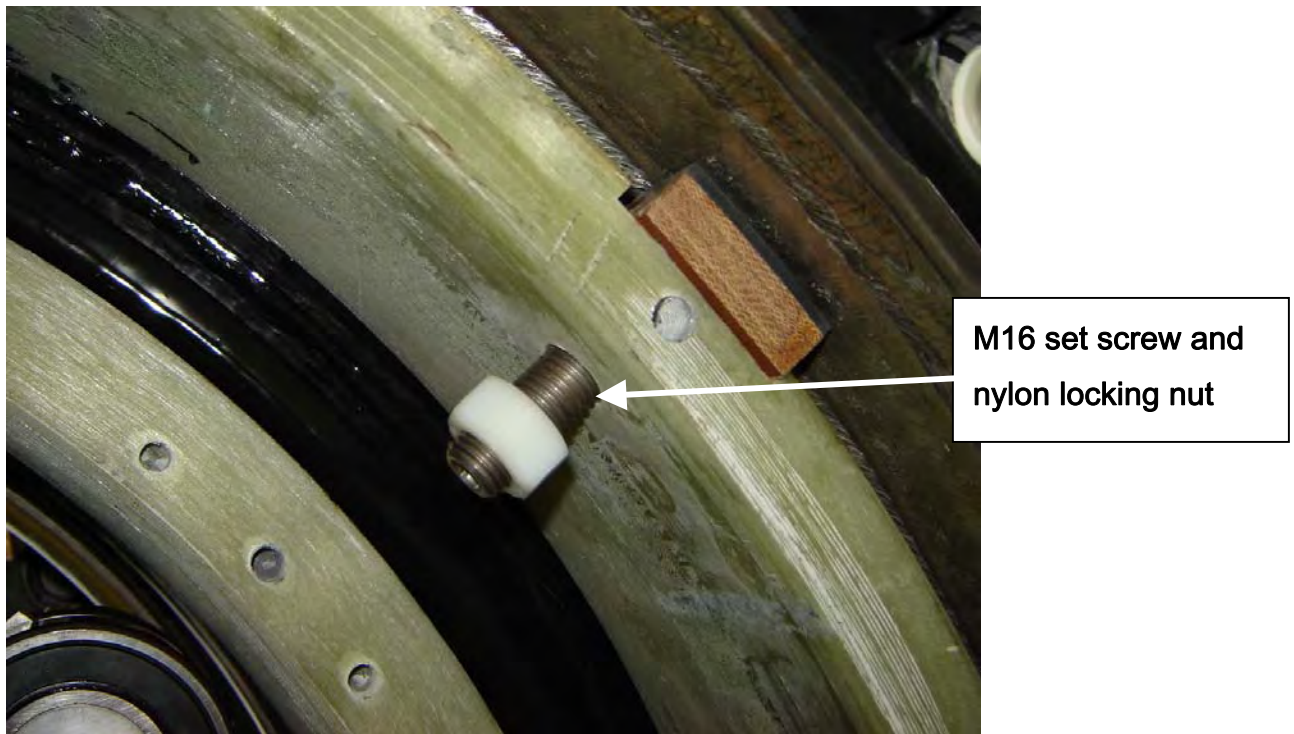
Illustration 14: Location Of Foam Filler Pads & Upper Radial Supports



NOTE: See Illustration 15 for an enlarged view of an upper radial support.

14. Pull out ½" black rubber cord or yellow foam from between gradient and magnet bore down to the gradient support bracket and discard.
15. Insert Gray Foam (Part Number 2379255-2) between the magnet warm bore and the outside diameter of the gradient coil on the service end of the magnet only. Cut foam to lengths, fitting between radial supports and gradient coil support on either side and between radial supports at the top.
16. For BRM / CRM install M16 Set screw (2293525) in the 2 positions indicated in Illustration 18; These will be installed at both ends of the gradient coil.
17. For BRM / CRM install mounting pad part number 2131953-4 with rubber against magnet ID (inner diameter) as shown in Illustration 15

Illustration 15: Enlarged View Of Upper Radial Support



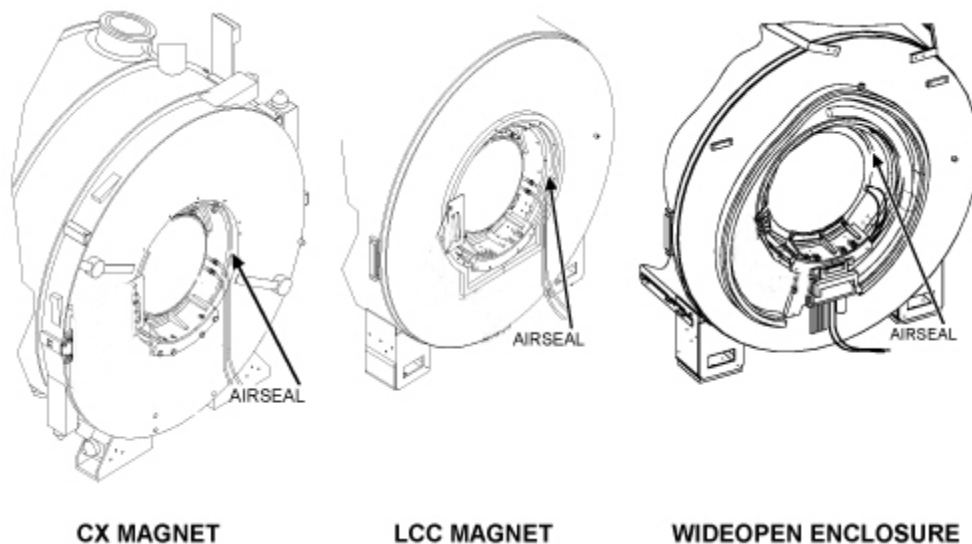
18. Tighten M16 stainless screws until snug (mounting pad doesn't move).
19. Then tighten each M16 set screw an additional $\frac{1}{2}$ turn.
20. Secure M16 set screw from loosening by jamming the nylon nut snug against the gradient coil (Nut contacts coil plus an additional $\frac{1}{2}$ turn).

NOTE: The gradient coil must have a uniform gap (between the outer diameter of the gradient coil and the inner diameter of the magnet warm bore). The coil can be centered using the mounting pad. Insert a mounting pad on the side of the gradient coil with the smallest gap. Tighten the cap screw until the coil is centered, then proceed inserting the remaining mounting pads (two on each side). After each of the remaining 3 mounting pads are in the snug position, Repeat steps 18-20

6. System Restoration

1. Replace the rear bottom RF Coil adjusting blocks.
2. Replace the poron seal gray pads to cover the 8 boltholes where the coil Interface bracket was installed.
3. Replace the Terminal Block or cable bracket located on the rear flange of the magnet.
(The Terminal Block is not used with the BRM-D and MR/i & CV/i WideOpen systems.)
4. Replace Open Foam Caulking Strip (2179397-3) and all Air Seals. See [Illustration 20](#).

Illustration 20: Replace Airseal



5. Confirm Gradient cooling air is not leaking.
6. Replace or restore the following:
 - o Signa Horizon Enclosure
 - Rear Enclosure Cover Replacement
 - Front Enclosure Cover Replacement
 - Patient Comfort Module Replacement

- Bridge and Cradle Replacement
- Rear Pedestal Replacement
- Signa MR/i & CV/i WideOpen Enclosure
 - Top, Bottom Trim , & Side Arc Cover Replacement
 - Rear Pedestal Cover Replacement
 - Bridge and Cradle Replacement
 - Rear Pedestal Replacement
 - Rear Endbell Replacement
 - Front Endbell Replacement

7. Perform Full SPT.

7. Finalization

Run EPI White Pixel Test.